



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
13-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**BEWARE OF NEGATIVE MARKING****PART-1 : PHYSICS****SECTION-I (i) : (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Four pair of dots are painted on a wall. Colour of pair of dots are red, green, blue and yellow. All are separated by equal distance. Which of the following pair of dots will appear to merge first as the observer starts moving away from the wall ?

- Red
- Green
- Blue
- Yellow

(Assume the person does not have colour blindness)

(A) Red

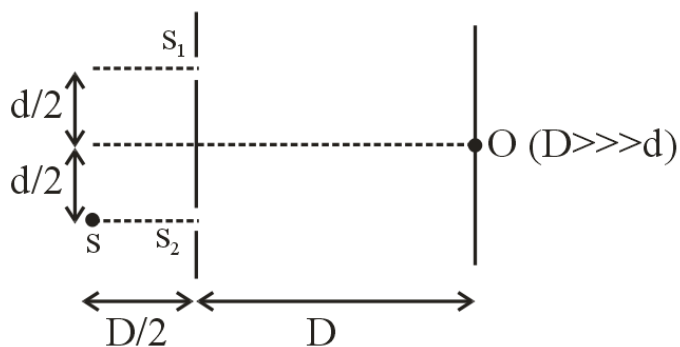
(B) Green

(C) Blue

(D) Yellow

Space for Rough Work

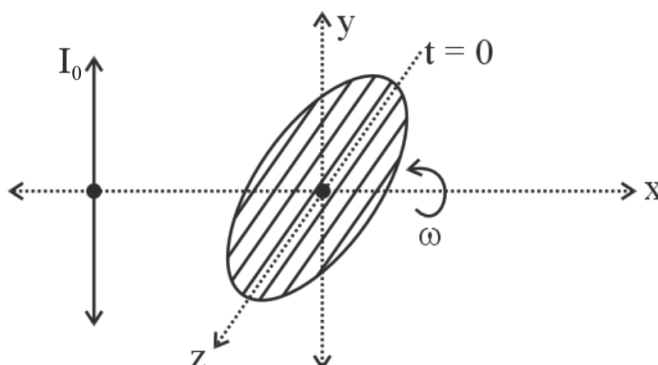
2. In the given diagram, find location and thickness of glass slab of refractive index ' μ ', such that central max. is at 'O'.



- (A) In front of ' S_1 ' $\frac{d^2}{2(\mu - 1)D}$
- (B) In front of ' S_2 ' $\frac{d^2}{(\mu - 1)D}$
- (C) In front of ' S_1 ' $\frac{d^2}{(\mu - 1)}D$
- (D) In front of ' S_2 ' $\frac{d^2}{2(\mu - 1)D}$

Space for Rough Work

3. In the given situation, electric field component of light of intensity ' I_0 ' is along y-axis and it is incident on a polariser kept in front of it in y-z plane. Initially pass axis of polariser is along z-axis and the polariser rotates with constant angular velocity ' ω ' with x-axis as rotation axis. Find the ratio of maximum rate of change in output intensity and angular frequency of rate of change in output intensity.



- (A) $\frac{I_0}{2}$ (B) $\frac{I_0}{8\omega}$ (C) $\frac{I_0}{8}$ (D) $\frac{I_0}{2\omega}$
4. An electro-magnetic wave of high intensity, has magnetic field amplitude ' B_0 '. It is incident on a small plane mirror of mass ' m ' and cross-section area ' A ' perpendicular to the plane of mirror, hanging from ceiling by a mass-less thread. The mirror is in equilibrium, then find tension in thread.

- (A) $\sqrt{(mg)^2 + \left(\frac{B_0^2 A}{2\mu_0}\right)^2}$ (B) $\sqrt{(mg)^2 + \left(\frac{B_0^2 A}{\mu_0}\right)^2}$
- (C) $\sqrt{(mg)^2 + \left(\frac{B_0^2}{2A\mu_0}\right)^2}$ (D) $\sqrt{(mg)^2 + \left(\frac{B^2}{A\mu_0}\right)^2}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

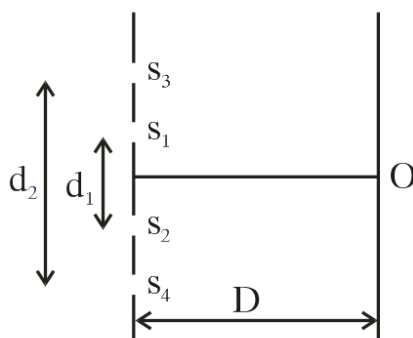
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

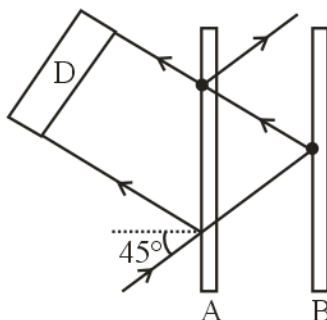
5. In an experiment of interference, there are two pairs of slits. Pair 1 (S_1 and S_2) have separation of $d_1 = 3$ mm. Pair 2 (S_3 and S_4) have separation of $d_2 = 4$ mm. Separation of slits from screen is $D = 1$ m. O is point on screen symmetrical to slits. Parallel beam of light perpendicular to plane of slits of wavelength $\lambda_1 = 400$ nm is incident on pair 2 and $\lambda_2 = 600$ nm is incident on pair 1. Both wavelengths λ_1 and λ_2 forms central maxima at O and central maxima of individual pair has intensity I_0 and $2I_0$ respectively. Which of the following is/are correct ?



- (A) Minimum intensity on screen is I_0
- (B) Minimum intensity on screen is $\frac{5I_0}{8}$
- (C) Minimum separation between point of minimum intensity is $66.67 \mu\text{m}$
- (D) Minimum separation between two point of minimum intensity is $200 \mu\text{m}$.

Space for Rough Work

6. A parallel beam of monochromatic microwave radiation of intensity I_0 is incident on a thin glass plate A as shown in figure. Another identical glass plate B is kept close to the first one and parallel to it. Width of the glass plate is $2\mu\text{m}$. Each glass plate reflects 25% of the light incident on it and transmit the remaining. Separation between the slabs is $\sqrt{2}\text{ cm}$. [Consider microwave frequency range to be 1 GHz - 100 GHz in general]



A detector is placed on the left of plate A. It receives radiation of 1st reflection from plate A and transmitted radiation after 1st reflection at B.

- (A) The detector receives maximum intensity if the frequency of microwave is 30 GHz.
 (B) The detector receives maximum intensity if the frequency of microwave is 15 GHz.
 (C) The maximum intensity received possibly by detector would be $\frac{49}{64}I_0$
 (D) The minimum intensity received possibly by detector would be $\frac{I_0}{64}$
7. Fundamental frequency of a stretched sonometer wire is f_0 . When its tension is increased by 96% and length decreased by 30%, its fundamental frequency becomes xf_0 . When its tension is decreased by 36% and its length is increased by 20%, its fundamental frequency becomes yf_0 , then (in each case mass/length remains same)

- (A) $\frac{x}{y} = 3$ (B) $\frac{x}{y} = \frac{5}{4}$ (C) $xy = \frac{5}{3}$ (D) $xy = \frac{4}{3}$

Space for Rough Work

8. A wire is stretched between two fixed ends 5 m apart. The extension in the wire is 12.5 mm and it's Young's modulus is $45 \times 10^{10} \text{ N/m}^2$. If density of wire is $4.5 \times 10^3 \text{ kg/m}^3$, then
- (A) Stress in the wire is $1.125 \times 10^7 \text{ W/m}^2$
- (B) Frequency of second overtone is 150 Hz
- (C) Frequency of fundamental tone is 50 Hz
- (D) Frequency of 2nd overtone is 100 Hz
9. The equation of a sound wave generated by a point source is given as $y = A \sin(15\pi x - 1500\pi t)$. An observer is standing at 10 m from source. If both source and observer are in medium of density 0.5 kg/m^3 and observer hears a pressure amplitude of 50 Pa, then
- (A) Bulk's modulus of medium is $5 \times 10^3 \text{ N/m}^3$
- (B) Displacement amplitude is $\frac{1}{15\pi} \times 10^{-2} \text{ m}$
- (C) Intensity received by observer is 25 W/m^2
- (D) Velocity of sound wave is 100 m/s.
10. A resonance column apparatus has pipe of radius 5 cm and length 1 m. If a tuning fork of frequency 540 Hz is used to produce standing waves, what can be the length of water column for resonance ? Speed of sound = 324 m/s.
- (A) 88 cm (B) 28 cm
- (C) 85 cm (D) 52 cm

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

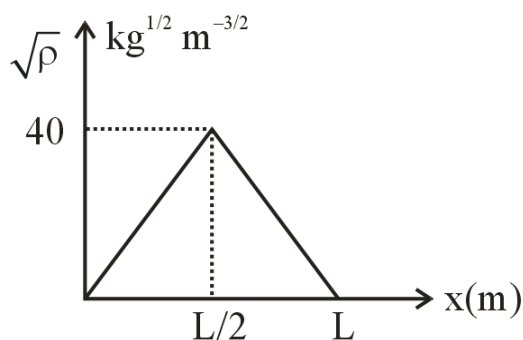
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Density of a non uniform wire varies from one end to another end as shown. Tension in the wire is uniform and equal to 10 N and area of cross section is 10 mm^2 . Length of wire is 10 m.



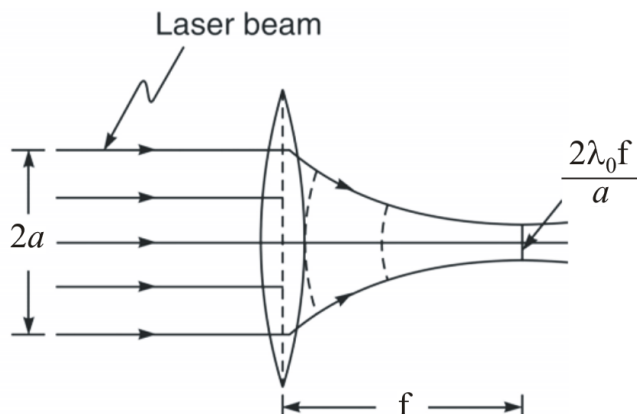
1. Calculate the mass of wire in gram.
2. A pulse is created in the wire at one of the end ($x = 0$). How much time (in ms) it takes for pulse to travel from $x = 2\text{m}$ to $x = 6\text{ m}$?

Space for Rough Work

Paragraph for Questions 3 and 4

A beam is said to be 'diffraction limited' if it diverges only due to diffraction. On the other hand, if we have tiny filament at the focal plane of a lens, the beam will primarily diverge due to finite size of filament.

Laser beams are usually diffraction limited.



As shown in above figure when a laser beam of diameter $2a$ incidents on a lens of focal length f , the waves will get focused to a spot of diameter $\frac{2\lambda_0 f}{a}$. Consider a 2 mW laser of diameter 1 mm which is incident on the eye. Focal length of pupil is 2.5 cm and wavelength of light 400 nm.

3. If area of focused spot is $x \times 10^{-5} \text{ mm}^2$, then calculate x . (Take $\pi = 3.14$)
4. If intensity of light on retina is $y \times 10^6 \text{ W/m}^2$ then, y is _____. such a high intensity can damage retina permanently.

Space for Rough Work

Paragraph for Questions 5 and 6

When unpolarised light passes through a polariser it allows the electric field oscillation parallel to transmission axis and stops oscillation perpendicular to it. This is possible only if polariser has sufficient layers of molecule. If thickness of film is inadequate, it is not able to polarise the light completely.

One such filter is able to reduce intensity of electric field to half of initial value in the direction perpendicular to transmission axis. Incident unpolarised light ray has intensity of 120 W/m^2 .

5. What will be intensity of the light (in W/m^2) after passing through such a faulty filter ?
6. One more such filter is placed after first one. It has its transmission axis perpendicular to first then, what will be intensity of light in W/m^2 after passing through both filter ?

Space for Rough Work

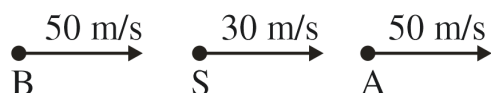
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. The speed of sound in a sample of hydrogen at 0°C is 1200 m/s. Some amount of oxygen gas is introduced in the vessel which causes reduction of speed of sound by half. Now an electric arc is given to the gases in the vessel causing formation of water vapour consuming all the oxygen. Assuming the water vapour to be ideal gas and temperature to be same as before, the speed of sound in the vessel now becomes $\sqrt{k}$ times 120 m/s. Find k after rounding off to nearest integer.
8. A sound source 'S' is moving with velocity 30 m/s and two observers A and B are moving with velocity 50 m/s as shown in figure. Velocity of sound 330 m/s. If the ratio of wavelength received by A and B is $\frac{5}{n}$, then n is equal to :



9. A parallel plate capacitor has circular plates with radius 50 cm and spacing 1.0 mm. A uniform electric field between the plates is changing at the rate of 1×10^6 V/ms. The magnetic field between the plates 15 cm from the axis is given by $B_0 \times 10^{-13}$ T. Find the value of B_0 after rounding off to nearest integer.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. The correct set of the solid and its respective crystal system is :

- (A) ZnS (Zinc blend) : Hexagonal
 - (B) Tin (White) : Tetragonal
 - (C) Sulphur (Rhombic) : Rhombohedral
 - (D) CaCO_3 (Calcite) : Triclinic
-

Space for Rough Work

2. Which of the following is true according to the ideal gas laws :
- (A) 2ℓ of a fixed moles of an ideal gas at 20°C when isobarically heated to double its temperature to 40°C , makes its volume double to 4ℓ .
- (B) 2ℓ of a fixed moles of an ideal gas at 20°C when isobarically heated by 1°C its volume increases by $\frac{1}{273.15} \times 2\ell$.
- (C) Both (A) & (B) are wrong.
- (D) Both (A) & (B) are correct.
3. The pH of a solution of 2ℓ $2\text{M H}_2\text{SO}_4$ when diluted to 20 litres is [use $\log_{10}2 = 0.3$]
- (A) 0.4 (B) 0.2 (C) 0.1 (D) 0.6
4. Which of the following is true for a zero order chemical reaction ?
- (A) It is always a multi-step reaction
- (B) The rate of a reaction increases on decreasing the concentration of the reactants.
- (C) Hydrogenation of ethene is an example of zero order reaction
- (D) The rate of reaction remains constant with temperature.
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

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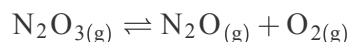
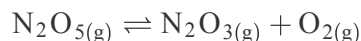
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Which is/are true :

- (A) A $3d_{z^2}$ orbital has zero electron density in the xy-plane
 - (B) Only three quantum numbers are needed to describe an orbital in an atom.
 - (C) The valence shell electronic configurations of all the elements of a given group of the periodic table is always similar.
 - (D) All d-orbital have two angular nodes.
-

Space for Rough Work

6. Consider the two simultaneous equilibria being established in a closed rigid vessel.



Upon addition of a little moles of $\text{N}_2\text{O}_{5(g)}$ to the equilibrium mixture, what is/are true with respect to the new equilibrium concentrations of the chemical species with respect to their concentration at the initial equilibrium situation :

- (A) N_2O_5 decreases (B) N_2O_3 increases
 (C) N_2O decreases (D) O_2 increases
7. An ionic crystal A^+B^- crystallises in NaCl type lattice, such that the shortest distance between A^+ and B^- ions is $3\text{\AA}$. Then which is/are true :
- (A) Under very high pressure, this NaCl type crystals change into the CsCl type crystals such that the packing efficiency changes.
 (B) Assuming the ideal radius ratio for the ions in the NaCl type crystal, the distance between the nearest cations is the same as that of the distance between the nearest anions.
 (C) Assuming the ideal radius ratio for the ions in the NaCl type crystal, the distance between the 2D-HCP layers of B^- ions is $2\sqrt{3}\text{\AA}$.
 (D) All the cations are present in a void with co-ordination number of 8.

Space for Rough Work

8. Consider two similar rigid containers at 300 K. One of them has 1 mole He(g) [4 amu] in it and the other one has 1 mole Ne(g) [20 amu]. If radius of He atom = 140 pm and radius of Ne atom = 154 pm, then which of the following is/are true.
- (A) The mean free path of He is more than that of Ne
 (B) The average speed of He is more than that of Ne
 (C) The collision frequency of He is more than that of Ne
 (D) Upon increasing the temperature, the mean free path for both He and Ne decrease.
9. Which of the following aqueous solution(s) is/are basic ?
- (A) 0.1 M NH_4Cl (B) 0.1 M KCN
 (C) 0.1 M $\text{CH}_3\text{COONH}_4$ (D) 0.1 M NaHCO_3
10. For the reaction : $\text{A} \rightarrow \text{B}$, the rate is $r = -\frac{d[\text{A}]}{dt} = k[\text{A}]^{-1}$. If the initial concentration of 'A' is A_0 M and its concentration at time 't' is A_t M, then identify the correct option(s) :
- (A) $A_t^2 = A_0^2 - 2kt$
 (B) $t_{50\%} = \frac{3A_0^2}{4k}$
 (C) $t_{100\%} = \frac{A_0^2}{2k}$
 (D) The reaction must be a complex reaction.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
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Paragraph for Questions 1 and 2

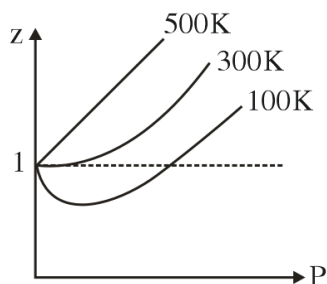
The emission spectrum is obtained when an excited sample of chemical de-excited and emits electromagnetic radiations.

1. Find the number of spectral lines obtained when a sample containing H atoms and He^+ ions both in $n = 4$ de-excite to ground level.
2. A hydrogen atom sample in its ground state is hit by photoelectrons ejected from a metal plate of work function 2eV . The photoelectrons were ejected when photons of energy $x\text{ eV}$ were incident on the metal plate. If the number of spectral lines obtained from the hydrogen sample were 3, then find x , considering the quantum efficiency for photoelectric effect to be 100%.

Space for Rough Work

Paragraph for Questions 3 and 4

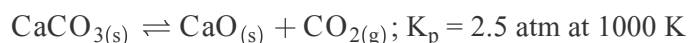
The compressibility factor (Z) for a real gas (Vander Waal) was plotted against pressure for a fixed moles of gas at different temperature as shown.



3. The critical temperature of this gas (in kelvin) is :
4. At 300 K, the gas behave like ideal gas upto 50 atm pressure. The molar volume (in L/mol) of gas at 300 K and 41 atm pressure, is ($R = 0.082 \text{ L-atm / K-mol}$)

Paragraph for Questions 5 and 6

Consider a reversible reaction.



[Use : $R = \frac{1}{12} \text{ L atm mol}^{-1} \text{ K}^{-1}$]

5. At 1000 K, a 100 litre of open rigid vessel was filled with 150 gram of CaCO_3 and then immediately covered and sealed tightly with a lid. The total pressure inside the container after a long time was P_0 atm. Find P_0 .
6. To the container [as described in the previous problem] a little $\text{CaCO}_3(s)$ is added and resealed again immediately. Now, the total pressure measured after a long time was P_1 atm.

Mark your answer as

- 1.11 if $P_1 > P_0$
- 2.22 if $P_1 < P_0$
- 3.33 if $P_1 = P_0$
- 4.44 if relation between P_1 & P_0 is ambiguous

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

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- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. The rate constant of a reaction at 27°C is $3 \times 10^{-3} \text{ min}^{-1}$. The fraction of molecule crossing the energy barrier at that temperature is $10^{-3}\%$. If temperature is increased, the maximum value of rate constant K can be attained is $x \times 100 \text{ min}^{-1}$. The value of x is :
8. 1500 ml 0.1 M H_3PO_4 is added to 500 mL 0.1 M Na_3PO_4 . Find the pH.
 Given, for H_3PO_4 : $\text{pK}_{a_1} = 2.5$, $\text{pK}_{a_2} = 7.5$ and $\text{pK}_{a_3} = 12.5$ at the given temperature.
 Use : $\log 2 = 0.3$, $\log 3 = 0.5$, $\log 5 = 0.7$
9. If the density of vapour of a substance (x) at 1 atm pressure & 500 K temperature is 0.8 kg/m^3 . The vapour effuses through a small hole at a rate of $\frac{4}{5}$ times slower than oxygen molecules, at same condition.
 What is the compressibility factor of the vapour ? (take $R = \frac{1}{12} \text{ L atm K}^{-1} \text{ mol}^{-1}$)
 (Express your answer after multiplying by 100).
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Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $\int \frac{\sin 7\theta}{A} d\theta = f(\theta) + C$, where $A = \prod_{k=1}^6 \left(\cos \theta - \cos \frac{k\pi}{7} \right)$, then $f(\theta)$ can be
 (A) $\sin \theta$ (B) $2^7 \cos \theta$ (C) $-2^6 \cos \theta$ (D) None of these
2. $\int_0^{\infty} \frac{\sin(x - \frac{1}{x})}{x} dx$ is equal to
 (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) 1 (D) 0
3. If $(x - y) f(x + y) - (x + y) f(x - y) = 4xy (x^2 - y^2)$. then the smaller area common to both the curves is
 $(f(y))^{\frac{2}{3}} + (f(x))^{\frac{1}{3}} = 0$ and $(f(y))^{\frac{2}{3}} + (f(x))^{\frac{2}{3}} = 12$
 (A) $2\pi + \sqrt{3}$ sq. unit (B) $2\pi - \sqrt{3}$ sq. unit
 (C) $\pi + \sqrt{3}$ sq. unit (D) $\pi - \sqrt{3}$ sq. unit
4. If the general solution of differential equation $2yy' = e^{\left(\frac{x^2+y^2}{x}\right)} + \frac{x^2+y^2}{x} - 2x$ is $cx = \frac{1}{e^{f(x)}}$ (where c is constant of integration). Then $f(x)$ is
 (A) $e^{\frac{x^2+y^2}{x}}$ (B) $e^{-\frac{x^2+y^2}{x}}$ (C) $e^{(x^2+y)}$ (D) $e^{-(x^2+y^2)}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If $\int \frac{(3x^{10} + 2x^8 - 2) \sqrt[4]{x^{10} + x^8 + 1}}{x^6} dx = \frac{p}{q} \frac{(x^{pq} + x^{4p} + 1)^{q/2p}}{x^q} + C$, then (where, p & q are co-prime)

- (A) $p + q = 7$ (B) $q - 2p = 1$ (C) $p + 2q = 11$ (D) $q^p = 25$

6. Let $f(x) = \int_0^4 |t - x| dt$ then, for $f(x)$;

- (A) Number of solutions of $f(x) = \sin x$ is 0 (B) Number of solutions of $f(x) = 4 |\sin x|$ is 0
(C) Number of solutions of $f(x) = \sin x$ is 1 (D) Number of solutions of $|f(x)| = 4 \sin x$ is 2

Space for Rough Work

7. A rectangular plane having vertices (0, 0), (2, 0), (0, 3) and (2, 3) rotates clockwise 90° about point (2, 0), then 90° clockwise about (5, 0), then clockwise 90° about (7, 0) and finally 90° clockwise about (10, 0). The area bounded by the curve traced by the point initially located at (1, 1) with x-axis between ordinates $x = 1$ and $x = 11$.

(A) $6 + \frac{5\pi}{2}$

(B) $7 + 6\pi$

(C) $6 + \frac{7\pi}{2}$

(D) is $p + \frac{q\pi}{2}$, then $p + q = 13$

8. Let $I = \int_0^1 \frac{dx}{1+x^2}$, then

(A) $I > \ln 2$

(B) $I < \ln 2$

(C) $I > \frac{\pi}{4}$

(D) $I < \frac{\pi}{4}$

9. Which of the following is/are true ?

(A) If $y = f(x)$ is a strictly monotonic function in (a, b) with $f(x) \neq 0$, then the area bounded by $y = f(x)$, the lines $x = a$, $x = b$ and $y = f(c)$ where $c \in (a, b)$ is minimum when $c = \frac{a+b}{2}$

(B) If $y = f(x)$ is a strictly monotonic function in (a, b) with $f(x) \neq 0$, then the area bounded by $y = f(x)$, the lines $x = a$, $x = b$ and $y = f(c)$ where $c \in (a, b)$ is minimum when $c = \frac{2a+b}{3}$ or $c = \frac{a+2b}{3}$

(C) If the area bounded by $f(x) = \frac{x^3}{3} - x^2 + a$ and the lines $x = 0$, $x = 2$ and the x-axis is minimum when $a = \frac{2}{3}$

(D) If the area bounded by $f(x) = \frac{x^3}{3} - x^2 + a$ and the lines $x = 0$, $x = 2$ and the x-axis is minimum when $a = \frac{28}{81}$

10. Solution of the differential equation $\frac{dy}{dx} = \frac{3x^2y^4 + 2xy}{x^2 - 2x^3y^3}$ is/are

(A) $x^2y^3 + \frac{x}{y^2} = c$

(B) $x^3y^2 + \frac{x^2}{y} = c$

(C) $\frac{x^2}{y^3} + xy^2 = c$

(D) $x^3y^2 - \frac{x^2}{y} = c$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

If $y = \int_{u(x)}^{v(x)} f(t) dt$, Let us define $\frac{dy}{dx}$ in a different manner as $\frac{dy}{dx} = v'(x)f^2(v(x)) - u'(x)f^2(u(x))$. The

equation of tangent at (a, b) is given by $y - b = \left(\frac{dy}{dx} \right)_{(a,b)} (x - a)$. Also $\int_a^a f(t) dt = 0$.

1. If $y = \int_x^{x^2} t^2 dt$, then x-intercept of tangent at $x = 1$ is
2. If $F(x) = \int_1^x e^{\frac{t^2}{2}} (1 - t^2) dt$, then $\frac{d}{dx}(F(x))$ at $x = 1$ is

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Paragraph for Questions 3 and 4

Given a function 'g' which has a derivative $g'(x)$ for every real 'x' and which satisfy $g'(0) = 2$ and $g(x + y) = e^y \cdot g(x) + e^x \cdot g(y) \forall x, y$.

3. If $g(\pi) g(-\pi) = a\pi^2$ then $|a|$ is equal to
4. If area bounded by $g(x)$ and ordinate of it's minima and co-ordinate axis is $2 - \frac{b}{e}$, then b is equal to

Paragraph for Questions 5 and 6

Consider the function $f(x) = \{x\} \sin(\pi x)$ where $\{.\}$ is fractional part.

5. If the area bounded by the graph of $f(x)$, x-axis, $x = -3$ and $x = b$ is $\frac{10}{\pi}$ then the value of b is equal to
6. The area bounded by the curve $y = f(x)$, x axis, $x = -\frac{1}{2}$ and $x = \frac{5}{2}$ is A, then $[A]$ has value equal to ($[]$ is GIF)

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. If $x \in \left(0, \frac{\pi}{2}\right)$ satisfies the equation $x^2 \int_0^1 \frac{dt}{t^2 + 2t \sin x + 1} - x \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{t^2 \tan 3t}{t^2 + 5} dt - \frac{1}{4} = 0$ then number of values of x which satisfy the equation is
8. Consider the square with vertices $(1, 1)$, $(-1, 1)$, $(-1, -1)$ and $(1, -1)$. Let S be the region consisting of all points inside the square which is nearer to the origin than any side. The value of $[S]$ is (use : $\sqrt{2} = 1.4$) and $[.] = \text{GIF}$)
9. If $x \cos\left(\frac{y}{x}\right) (y dx + x dy) = y \sin\left(\frac{y}{x}\right) (x dy - y dx)$, $y(1) = 2\pi$, then the value of $4 \frac{y(4)}{\pi} \cos\left(\frac{y(4)}{4}\right)$ is equal to

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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